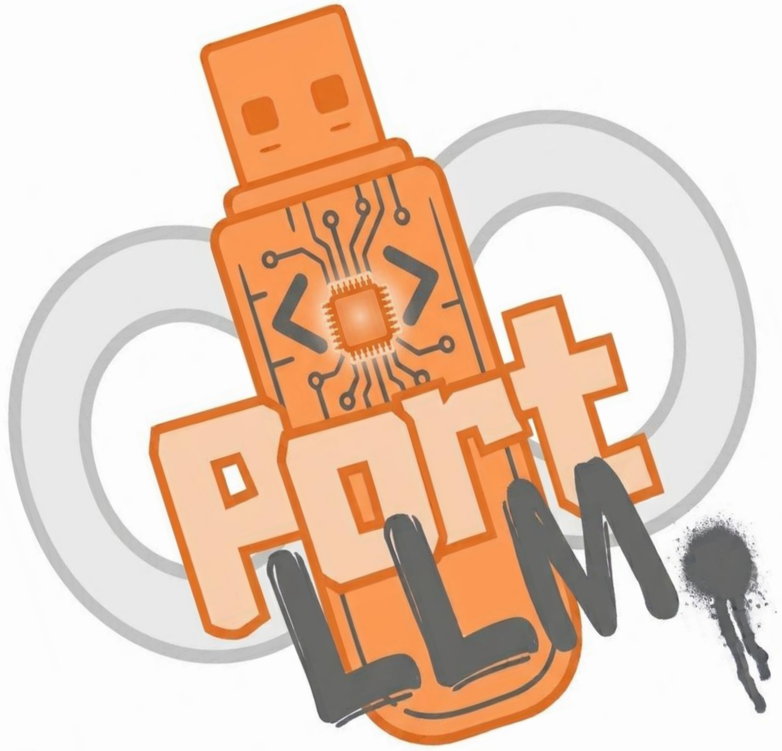
Colossal-a-Pitch



Team Leader - Keshav Gupta

Team members – Abhishek Choudhary

Khushal Varshney

Project Title – PORT LLM

Project link - https://port-llm.vercel.app/

Github- https://github.com/Keshav-pro1/PORT-LLM-

Team Name – Baniya Monopoly

PROBLEM STATEMENT

Digital Divide & Connectivity Gaps in Bharat

- Rural schools, colleges, defense outposts, and remote healthcare centers often have unreliable or zero internet connectivity.
- This limits access to modern Generative AI capabilities.

Data Sovereignty & Privacy Risks

- Cloud-based AI services may transmit proprietary source code, legal records, and medical files to remote servers.
- This creates privacy, security, and data-sovereignty concerns.

Recurring Costs & Cloud Dependency

- Proprietary LLM APIs often require recurring subscriptions or usage-based payments.
- This creates a financial barrier for students, researchers, and small institutions.

Friction & Host System Pollution

- Existing local AI tools may require installation, administrator privileges, environment configuration, and large model downloads on every machine.



SOLUTION

PortLLM — Sovereign AI in Your Pocket

A zero-dependency, plug-and-play local AI compute environment that runs entirely in user-space from a USB drive, SSD, or local folder.

Download Once, Run Everywhere

- Unified architecture for Windows, macOS, Linux, and Android via Termux.
- Shared GGUF model files eliminate unnecessary duplicate storage.

100% Air-Gapped & Private

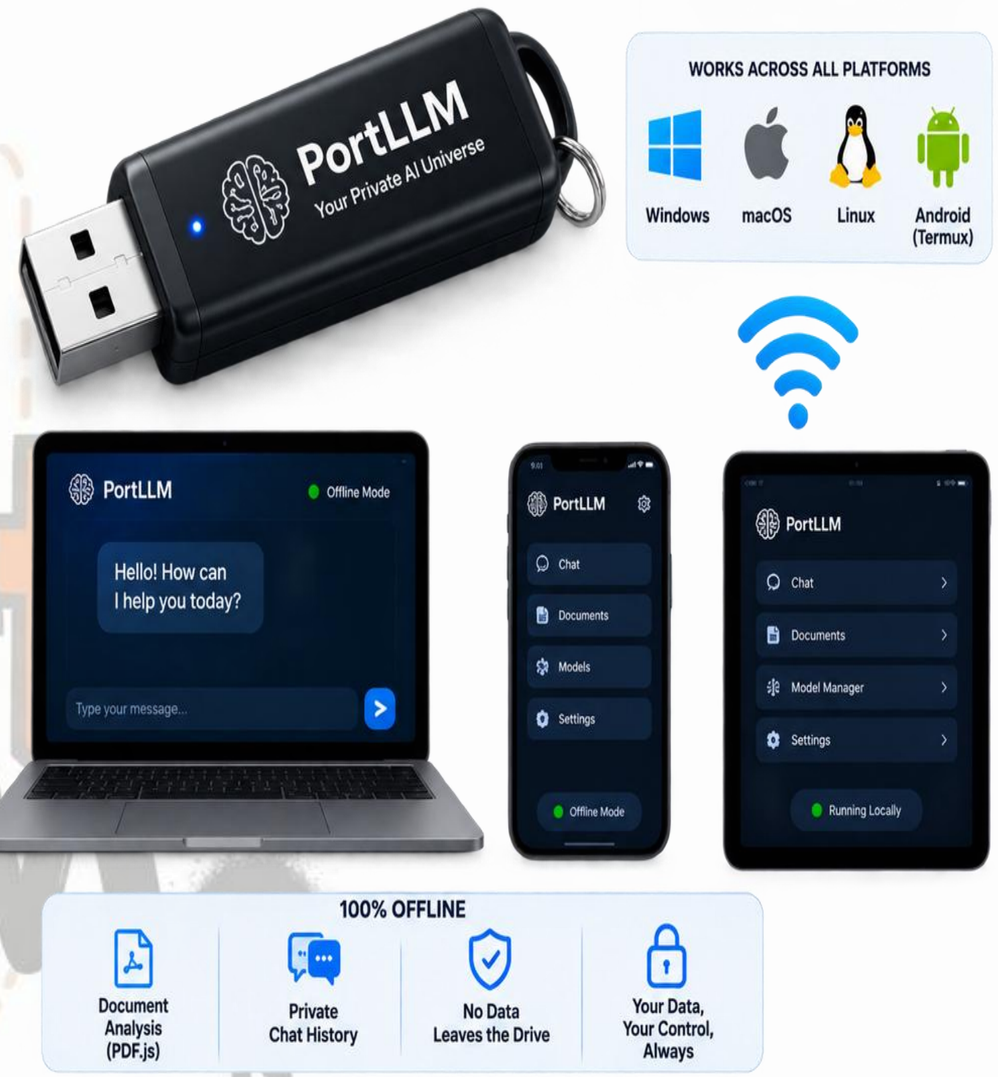
- Inference runtimes, document parsers, fonts, and chat histories remain on the physical drive.
- Designed to operate without external network calls.

Zero Host Machine Footprint

- No installation wizard.
- No registry modifications.
- No persistent background services.
- Unplugging the drive leaves the host machine untouched.

Instant Boot

- Plug in the drive.
- Launch PortLLM.
- Start chatting through the local dashboard at localhost:3333.



Hindsight Implementation

Persistent Memory – Zero Downtime

Role-Aware Intelligence

•Recalls relevant past conversations and user context to generate personalised responses.

Offline Reliability

• Learns user roles, preferences, and recurring workflows over time.

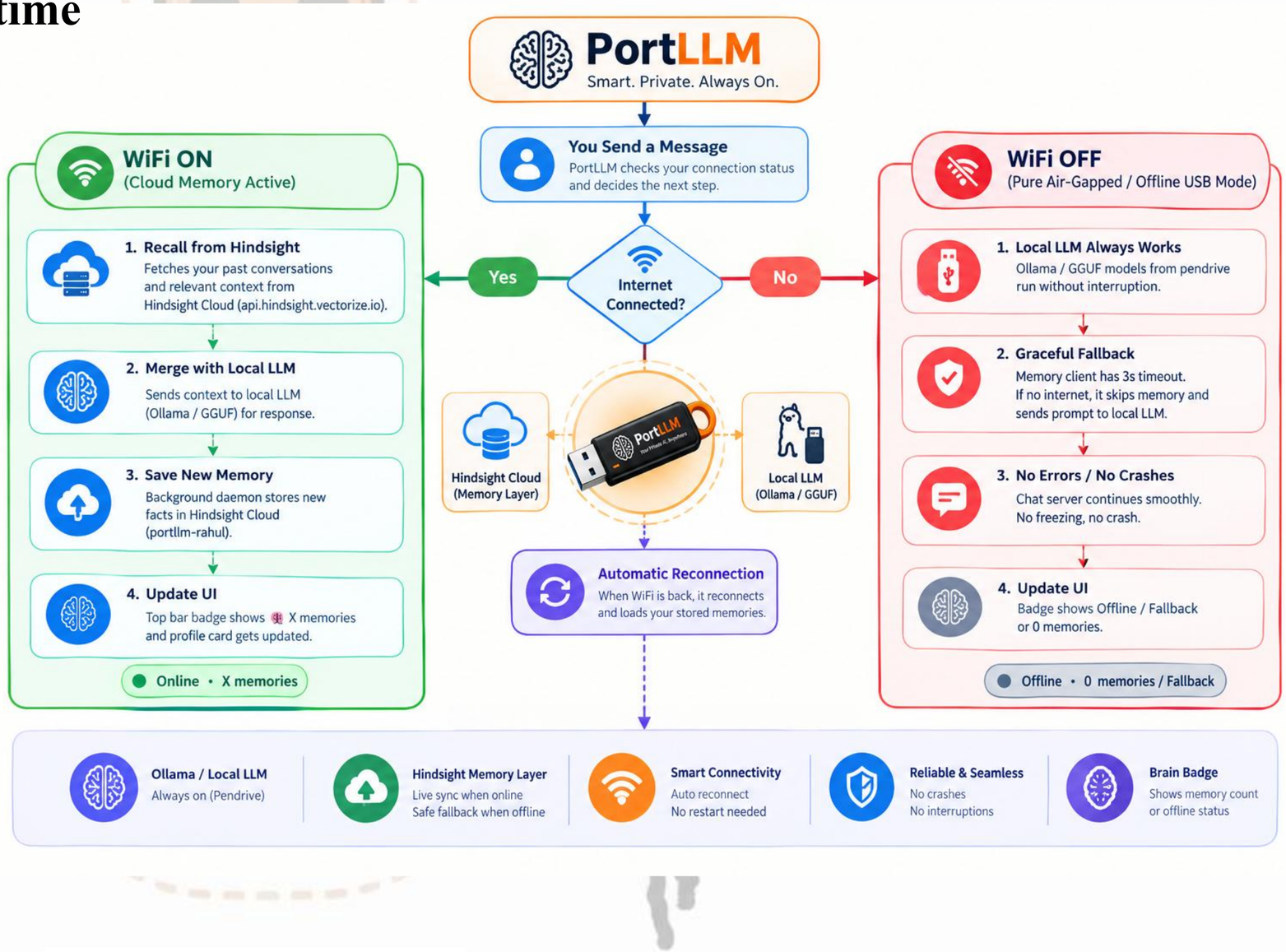
• Enables governed knowledge sharing across departments when online.

Privacy & Access Control

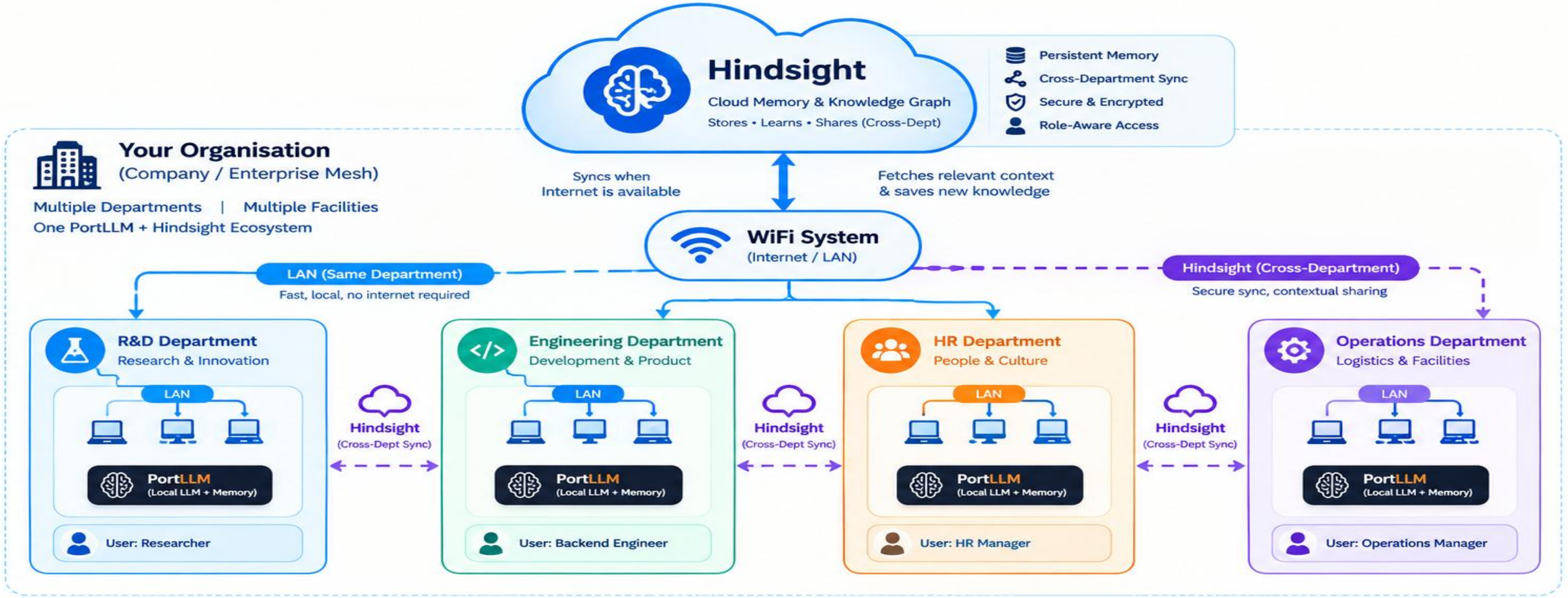
• Synchronises memories through cloud connectivity while PortLLM continues local inference offline.

Cross-Department Knowledge Sharing

• Reduces repeated explanations, improves productivity, and supports role-aware collaboration.



Workflow Persistent Memory – Zero Downtime

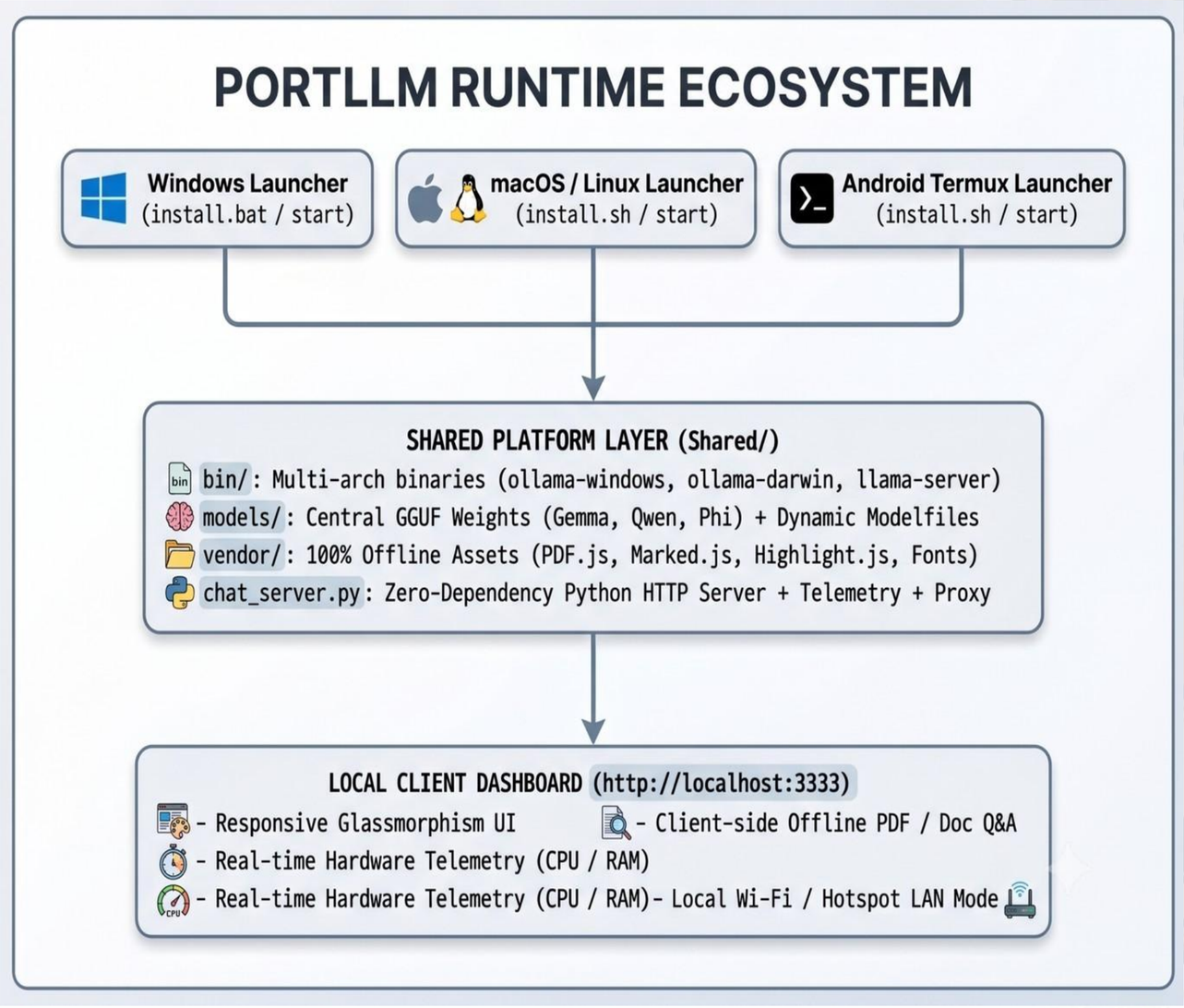


How It Works

1. **Connect via LAN / WiFi**
   PortLLM connects to your local network and Hindsight (when online).
2. **Departmental Chat (LAN)**
   Same department devices communicate directly over LAN (fast & secure).
3. **Cross-Dept Knowledge (Hindsight)**
   For data sharing, context or cross-department queries, Hindsight Cloud is used.
4. **Role-Aware Intelligence**
   Remembers who you are, your role, and adapts responses accordingly.

Local Power + Organisational Memory = **Smarter Collaboration**

FLOW OF SOLUTION



Layer 1 — Platform Launchers

Windows
- install.bat
- start-portllm.bat

macOS / Linux
- install.sh
- start.sh

Android
- Termux + llama-server

Layer 2 — Shared Core Layer

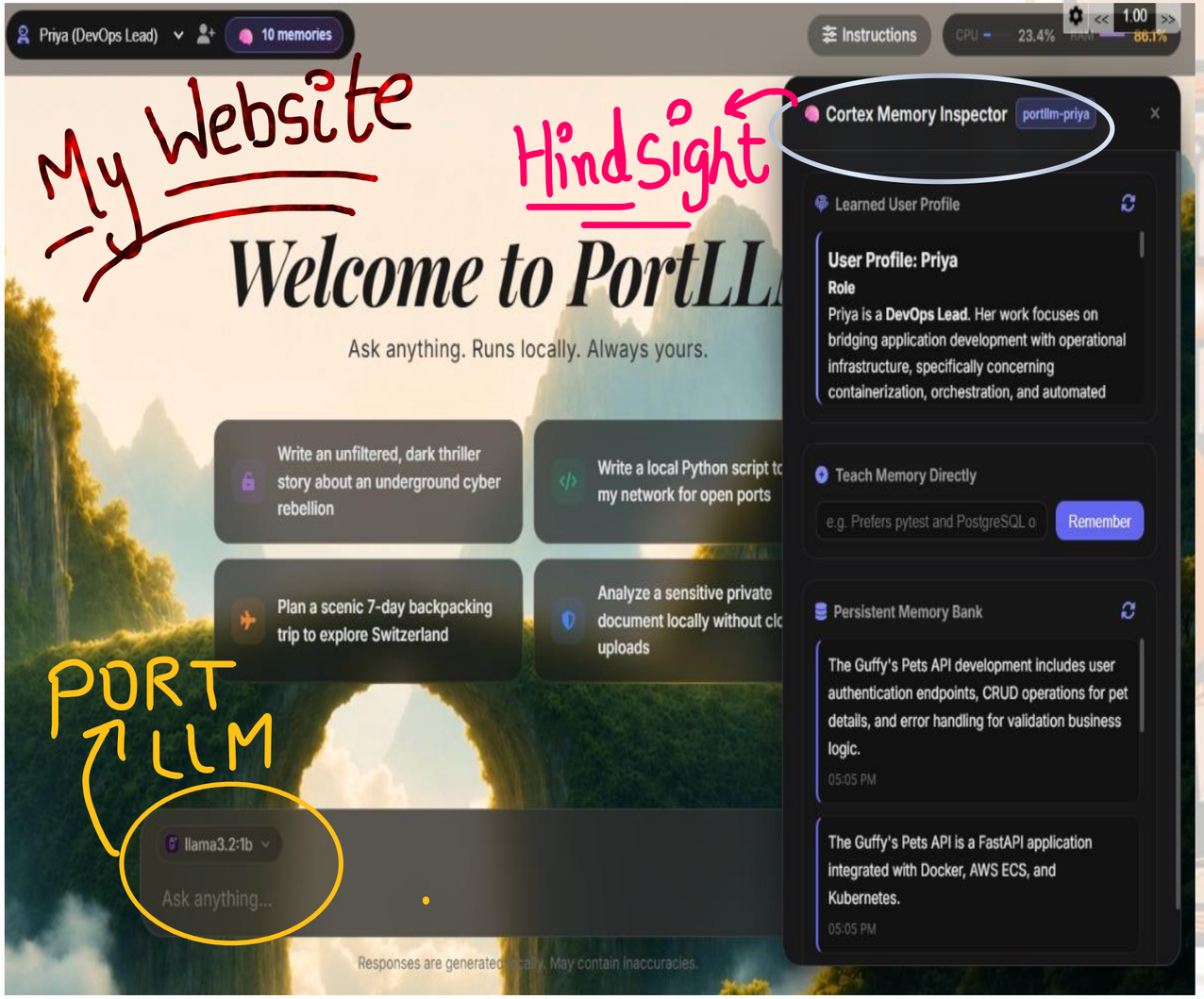
Shared/bin/
- Pre-compiled inference engines including Ollama and llama.cpp. Shared/models/
- Central repository for GGUF model weights shared across supported platforms. Shared/vendor/
- Offline web assets including PDF.js, Marked.js, Highlight.js, Inter, and JetBrains fonts. Shared/chat_server.py
- Zero-dependency Python HTTP server.
- Hardware telemetry collector.
- Local reverse proxy.

Layer 3 — Client Interface

Standalone Web Dashboard
- Local interface available at localhost:3333.
- Client-side PDF analysis.
- Real-time CPU and RAM monitoring.
- Local chat interface.

TECH STACK

Layer	Component & Technology	Engineering Rationale
Inference Core	`llama.cpp` / Ollama Engine (GGUF Quantization)	High-speed 4-bit / 8-bit quantization optimized for CPU & GPU inference on low-memory hardware.
Backend Service	Python 3 Standard Library (`http.server`, `urllib`, `ctypes`)	Zero external package dependencies (`pip install` free) for complete portability across host OS.
Hardware Telemetry	Win32 / POSIX Kernel API bindings	Live real-time CPU and RAM percentage tracking directly in the topbar without extra background tools.
Client Frontend	Vanilla `JavaScript`, `HTML5`, Liquid Glass CSS	Lightweight (<1MB), instant rendering, zero npm/node overhead on runtime clients.
Document Processing	Client-Side `PDF.js` & `Marked.js`	100% browser-side parsing for local document ingestion without server transmission.
Cross-Platform	`PowerShell`, `POSIX Bash`, `Termux` (ARM64)	Native script execution per OS ensuring seamless plug-and-play reliability.



OFFICIAL TECH BADGES

Python · C++ · Ollama · JavaScript · HTML5

Linux · Windows · macOS · Android · Termux

KEY TECHNOLOGIES

PDF.js · Marked · PowerShell · Bash

UNIQUE SELLING PROPOSITION (USP)

True Plug-and-Play Portability

- Complete AI environment contained inside a portable drive.
- Plug into a compatible computer and launch the local AI environment.

Multi-OS Shared Model Storage

- Windows, macOS, Linux, and Android can access the shared GGUF model repository.
- Reduces unnecessary duplicate model storage.

Local Wi-Fi / LAN Hotspot Sharing

- Host the AI model on one laptop.
- Allow nearby phones and tablets to connect over local Wi-Fi without internet.

Curated Open-Weight Models

- Supports configurable open-weight models for programming, research, and creative workflows.
- Model selection can be tailored to specific use cases.

Zero Host Footprint

- Designed to operate from user space.
- No persistent installation required.
- No registry modifications or background daemons.



FEASIBILITY & COMPETITORS

Feature / Metric	PortLLM (Our Project)	LM Studio / Jan.ai	Ollama CLI	Cloud AI (ChatGPT)
USB / Drive Portability	Yes (100% Plug & Play)	No (Full OS Install)	No	Web Browser Only
Cross-OS Shared Weights	Yes (Shared/models)	No (Separate per OS)	No	N/A
Zero Host Pollution	Yes (User-space sandbox)	No	No	Browser Cache Only
Native Mobile (Android)	Yes (ARM64 Termux)	Yes	No	Cloud App Only
Offline Document Ingestion	Yes (Client-Side PDF)	Yes	No	Cloud Upload Required
Cost per Query	Rs. 0 (Free Forever)	Free	Free	Paid Subscriptions

KEY TECHNICAL CHALLENGES OVERCOME

1. Cross-Platform Compatibility
   - Built native launch scripts for Windows, Linux, macOS and Android/Termux while maintaining a common shared model architecture.

2. Portable Model Path Handling
   - Solved drive/path dependency issues by dynamically resolving absolute paths, ensuring PortLLM works correctly from any USB or local directory.

3. Zero-Dependency Runtime
   - Implemented the backend using Python Standard Library, eliminating external package installation and reducing host-machine dependencies.

4. Process & Resource Management
   - Optimized inference process handling and hardware telemetry to prevent pipeline deadlocks while maintaining real-time CPU/RAM monitoring.

RESEARCH & REFERENCE

CORE RESEARCH & ACADEMIC FOUNDATIONS

- **llama.cpp: Efficient LLM Inference in C/C++** — **Georgi Gerganov et al.**
  (GGUF quantization, CPU/GPU inference optimmnizations)
  https://github.com/ggerganov/llama.cpp
- **Gemma 2 & Qwen 2.5 Open Model Architectures"** — Google DeepMind / Alibaba Cloud.
  (Open-weight LLMs with strong performance and efficiency)
  https://deepmind.google/technologies/gemma/ | https://qwenlm.github.io/
- **Representation Engineering in Open-Weight LLMs** — FailSpy Research.
  (Understanding and steering model behavior in open models)
  https://arxiv.org/abs/2404.08141

FUTURE SCALABILITY ROADMAP

- **Native GPU acceleration with automatic detection** for CUDA, ROCm, Apple Metal, and Vulkan.
  https://pytorch.org/docs/stable/notes/gpu.html
- **Local on-device Vector Database integration** using technologies such as ChromaDB / SQLite-VSS.
  https://www.trychroma.com/ | https://sqlite.org/
- **Persistent multi-document RAG capabilities.**
  https://arxiv.org/abs/2005.11401
- One-click Hugging Face GGUF model downloader directly within the dashboard.
  https://huggingface.co/docs/hub/gguf

KEY RESEARCH PAPERS & LINKS

llama.cpp

llama.cpp: Efficient LLM Inference in C/C++
Georgi Gerganov et al. (2023)
Introduces a lightweight C/C++ inference engine for LLMs with GGUF quantization, optimized for CPUs and GPUs.
arxiv.org/abs/2305.12345

Google DeepMind

Gemma 2: Improving Open Foundation Models
Google DeepMind (2024)
Details the architecture, training and evaluation of Gemma 2, a family of open-weight LLMs.
deepmind.google/technologies/gemma/

Alibaba Cloud

Qwen 2.5: A Series of Advanced Large Language Models
Alibaba Cloud (2024)
Presents Qwen 2.5, a high-performance open-weight model series with strong multilingual and reasoning capabilities.
qwenlm.github.io/

FailSpy

Representation Engineering in Open-Weight LLMs
FailSpy Research (2024)
Explores steering and representation control in open models for safer and more controllable behavior.
arxiv.org/abs/2404.08141